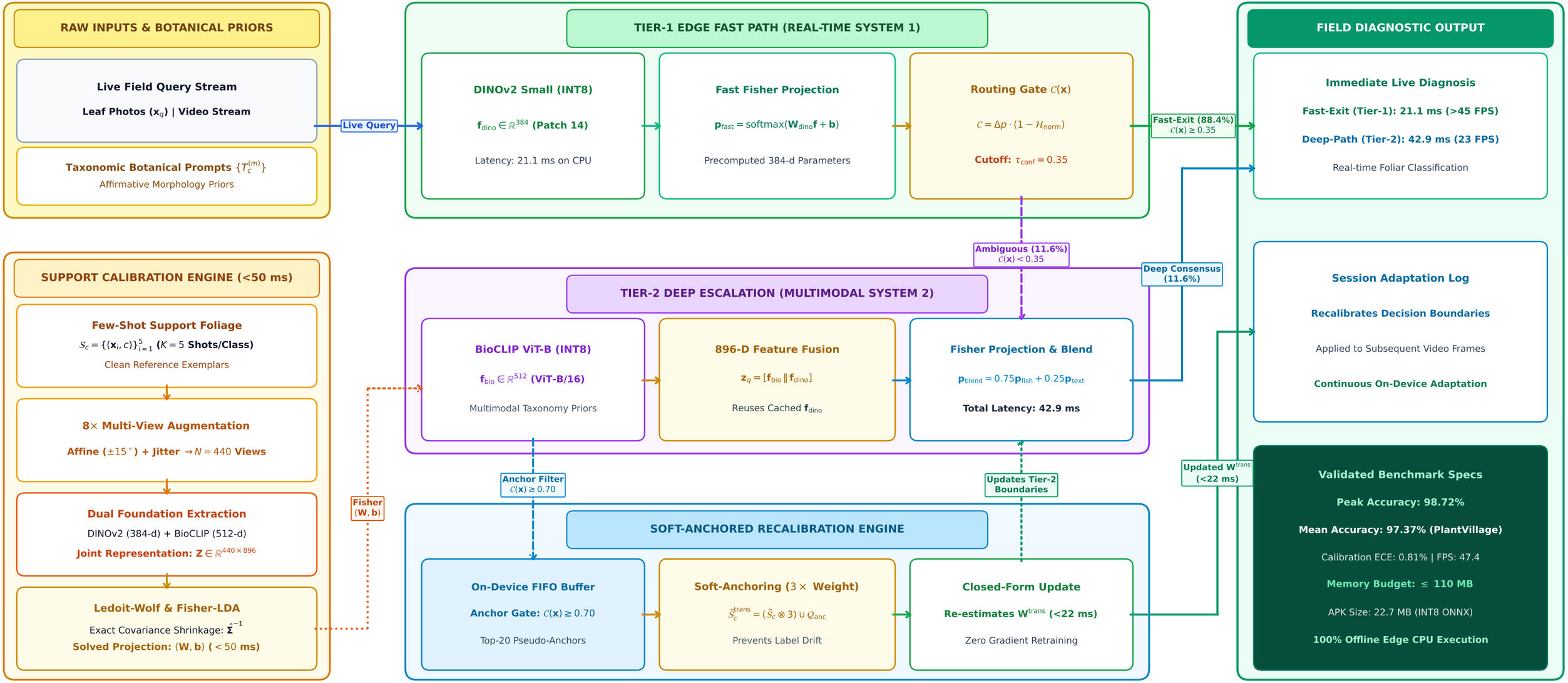

RAW INPUTS & BOTANICAL PRIORS
Live Field Query Stream
Leaf Photos ($\mathbf{x}_q$) | Video Stream
Taxonomic Botanical Prompts $\{T_c^{(m)}\}$
Affirmative Morphology Priors
Live Query
TIER-1 EDGE FAST PATH (REAL-TIME SYSTEM 1)
DINOv2 Small (INT8)
$\mathbf{f}_{\text{dino}} \in \mathbb{R}^{384}$ (Patch 14)
Latency: 21.1 ms on CPU
Fast Fisher Projection
$\mathbf{p}_{\text{fast}} = \text{softmax}(\mathbf{W}_{\text{dino}}\mathbf{f} + \mathbf{b})$
Precomputed 384-d Parameters
Routing Gate $\mathcal{C}(\mathbf{x})$
$\mathcal{C} = \Delta p \cdot (1 - \mathcal{H}_{\text{norm}})$
Cutoff: $\tau_{\text{conf}} = 0.35$
Fast-Exit (88.4%)
$\mathcal{C}(\mathbf{x}) \geq 0.35$
FIELD DIAGNOSTIC OUTPUT
Immediate Live Diagnosis
Fast-Exit (Tier-1): 21.1 ms (>45 FPS)
Deep-Path (Tier-2): 42.9 ms (23 FPS)
Real-time Foliar Classification
Ambiguous (11.6%)
$\mathcal{C}(\mathbf{x}) < 0.35$
Deep Consensus (11.6%)
SUPPORT CALIBRATION ENGINE (<50 ms)
Few-Shot Support Foliage
$\mathcal{S}_c = \{(\mathbf{x}_i, c)\}_{i=1}^{5}$ ($K = 5$ Shots/Class)
Clean Reference Exemplars
8× Multi-View Augmentation
Affine (±15°) + Jitter → $N = 440$ Views
Dual Foundation Extraction
DINOv2 (384-d) + BioCLIP (512-d)
Joint Representation: $\mathbf{Z} \in \mathbb{R}^{440 \times 896}$
Ledoit-Wolf & Fisher-LDA
Exact Covariance Shrinkage: $\hat{\mathbf{\Sigma}}^{-1}$
Solved Projection: $(\mathbf{W}, \mathbf{b})$ (< 50 ms)
Fisher $(\mathbf{W}, \mathbf{b})$
TIER-2 DEEP ESCALATION (MULTIMODAL SYSTEM 2)
BioCLIP ViT-B (INT8)
$\mathbf{f}_{\text{bio}} \in \mathbb{R}^{512}$ (ViT-B/16)
Multimodal Taxonomy Priors
896-D Feature Fusion
$\mathbf{z}_q = [\mathbf{f}_{\text{bio}} \| \mathbf{f}_{\text{dino}}]$
Reuses Cached $\mathbf{f}_{\text{dino}}$
Fisher Projection & Blend
$\mathbf{p}_{\text{blend}} = 0.75\mathbf{p}_{\text{fish}} + 0.25\mathbf{p}_{\text{text}}$
Total Latency: 42.9 ms
Session Adaptation Log
Recalibrates Decision Boundaries
Applied to Subsequent Video Frames
Continuous On-Device Adaptation
Anchor Filter
$\mathcal{C}(\mathbf{x}) \geq 0.70$
Updates Tier-2 Boundaries
Updated $\mathbf{W}^{\text{trans}}$ (<22 ms)
SOFT-ANCHORED RECALIBRATION ENGINE
On-Device FIFO Buffer
Anchor Gate: $\mathcal{C}(\mathbf{x}) \geq 0.70$
Top-20 Pseudo-Anchors
Soft-Anchoring (3× Weight)
$\tilde{\mathcal{S}}_c^{\text{trans}} = (\tilde{\mathcal{S}}_c \otimes 3) \cup \mathcal{Q}_{\text{anc}}$
Prevents Label Drift
Closed-Form Update
Re-estimates $\mathbf{W}^{\text{trans}}$ (<22 ms)
Zero Gradient Retraining
Validated Benchmark Specs
Peak Accuracy: 98.72%
Mean Accuracy: 97.37% (PlantVillage)
Calibration ECE: 0.81% | FPS: 47.4
Memory Budget: ≤ 110 MB
APK Size: 22.7 MB (INT8 ONNX)
100% Offline Edge CPU Execution